Proposed alternative method for creating 3D morphology, Letting the errors out of a morphology

In wall 4, the middle of each layer is steady, while the start forms a bump and the end falls into a loss ramp.

Option 1: use only the steady middle. Forms are built from the middle, and the start and end run out on tabs that are cut off after printing.

Option 2: vary wire feed speed (WFS) along the bead. Other papers also change the process along the bead: Lu et al. (2025), Wang et al. (2022), Banerjee et al. (2022) and Xiong and Zhang (2014). Our travel speed is fixed, so WFS is the knob we have. In 13 single beads, the extra metal at the start grew with WFS, from about 5 mm^3 at 150 in/min to about 32 mm^3 at 325 in/min.

Test plan. Treat each bead as three parts.

Start: vary WFS over the start part and measure how much it changes the bump.

Middle: keep schedule 22 (300 in/min).

End: measure how much material is missing at the end of each bead, with the settings used in steps 1 and 2. This shows whether a second pass over the end is worth testing.

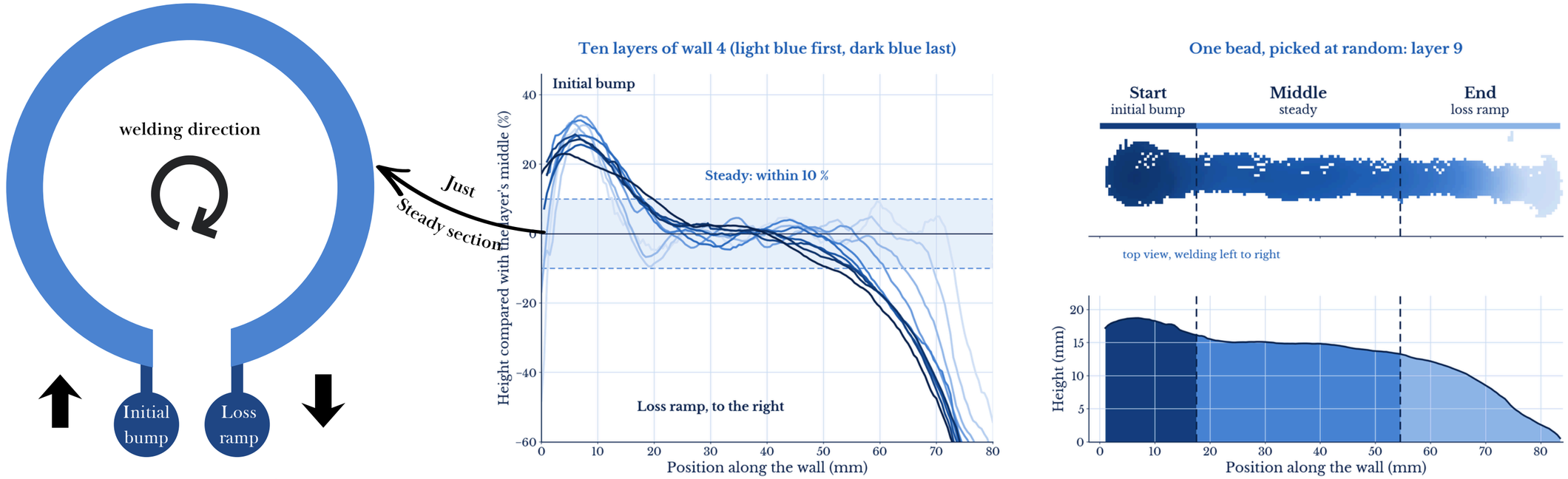


Figure. Possible implementation of a cylinder